

Real Analysis Lecture Notes

Sequences of Functions, Uniform Convergence, Differentiation/Integration,
Power Series, Abel's Theorem

Contents

1	Lecture Plan (Time Allocation)	3
2	Lecture 1: Pointwise and Uniform Convergence	4
2.1	Definitions	4
2.2	Uniform Convergence Implies Pointwise Convergence	4
2.3	Example: A Basic Uniform Limit Question	4
3	Lecture 2: Cauchy Criterion and Uniform Convergence Tests	5
3.1	Cauchy Criterion	5
3.2	Test Problems (Worked)	5
4	Lecture 3: Pointwise vs Uniform; When Pointwise $\Rightarrow$ Uniform	6
4.1	Classic Counterexample (Fully Worked)	6
4.2	When does pointwise convergence become uniform?	7
5	Lecture 4: Uniform Convergence and Differentiation	7
5.1	Main Theorem + Complete Proof	7
5.2	Application 1: $f_n(x) = \frac{\cos(nx)}{n}$	8
5.3	Application 2: $f_n(x) = x - \frac{x^n}{n}$ (Correct domain discussion)	9
6	Lecture 5: Uniform Convergence and Integration	9
6.1	Theorem + Complete Proof	9

6.2	Spikes: Why Uniform Matters (Full Computations)	10
7	Lecture 6: Cauchy Criterion Example $\tan^{-1}(nx)$	10
8	Lecture 7: Recursive Sequence $f_{n+1}(x) = \sqrt{x + f_n(x)}$	11
9	Lecture 8: Power Series, Radius of Convergence, Abel's Theorem	12
9.1	Radius of Convergence	12
9.2	Transforming the Radius	12
9.3	Computing Radii (Worked)	13
9.4	Abel's Theorem (Statements) and Applications	13

Source

These notes are compiled and expanded into full lectures (with complete proofs and worked solutions) from the question set in the PDF `Real-2.pdf`.

1 Lecture Plan (Time Allocation)

Assume each lecture is **90 minutes**. A suggested pacing:

- L1. (90 min)** Pointwise vs uniform convergence; uniform $\Rightarrow$ pointwise; key examples.
- L2. (90 min)** Cauchy criterion; testing uniform convergence by supremum estimates.
- L3. (90 min)** Pointwise but not uniform; when pointwise $\Rightarrow$ uniform (Dini, equicontinuity ideas).
- L4. (90 min)** Uniform convergence and differentiation: theorem + proof + applications.
- L5. (90 min)** Uniform convergence and integration: theorem + proof; counterexamples when uniform fails.
- L6. (90 min)** Advanced examples: spikes, oscillations; boundedness of derivatives vs uniform convergence.
- L7. (90 min)** Recursive sequences of functions $f_{n+1} = \sqrt{x + f_n(x)}$; uniform convergence via Dini.
- L8. (90 min)** Power series: radius of convergence; transformations; Abel's theorem + applications to $\log 2$ and $(\log(1+x))^2$.

2 Lecture 1: Pointwise and Uniform Convergence

2.1 Definitions

Definition 2.1 (Pointwise Convergence). Let $E \subseteq \mathbb{R}$ and $f_n : E \rightarrow \mathbb{R}$. We say $f_n \rightarrow f$ *pointwise* on E if for every $x \in E$ and every $\varepsilon > 0$, there exists $N = N(\varepsilon, x)$ such that

$$|f_n(x) - f(x)| < \varepsilon \quad \text{for all } n \geq N.$$

Definition 2.2 (Uniform Convergence). We say $f_n \rightarrow f$ *uniformly* on E if for every $\varepsilon > 0$, there exists $N = N(\varepsilon)$ such that

$$|f_n(x) - f(x)| < \varepsilon \quad \text{for all } x \in E \text{ and all } n \geq N.$$

Equivalently,

$$\sup_{x \in E} |f_n(x) - f(x)| \xrightarrow{n \rightarrow \infty} 0.$$

2.2 Uniform Convergence Implies Pointwise Convergence

Theorem 2.3. If $f_n \rightarrow f$ uniformly on E , then $f_n \rightarrow f$ pointwise on E .

Proof. Fix $x \in E$ and $\varepsilon > 0$. Uniform convergence gives an $N(\varepsilon)$ such that $|f_n(t) - f(t)| < \varepsilon$ for all $t \in E$ and all $n \geq N(\varepsilon)$. In particular, for this fixed x we have $|f_n(x) - f(x)| < \varepsilon$ for all $n \geq N(\varepsilon)$. Thus pointwise convergence holds at x . Since x was arbitrary, pointwise convergence holds on E . $\square$

2.3 Example: A Basic Uniform Limit Question

Example 2.4. Consider $f_n : (0, 1) \rightarrow \mathbb{R}$ given by

$$f_n(x) = \frac{nx}{n^2 + 1} \quad (n \geq 1).$$

Does it have a uniform limit? Find it and justify.

Solution. For each fixed $x \in (0, 1)$,

$$f_n(x) = x \cdot \frac{n}{n^2 + 1} \xrightarrow{n \rightarrow \infty} 0,$$

so the pointwise limit is $f(x) = 0$.

To test uniform convergence on $(0, 1)$, estimate the sup:

$$\sup_{x \in (0, 1)} |f_n(x) - 0| = \sup_{x \in (0, 1)} \frac{nx}{n^2 + 1} = \frac{n}{n^2 + 1} \sup_{x \in (0, 1)} x = \frac{n}{n^2 + 1}.$$

Since $\frac{n}{n^2 + 1} \rightarrow 0$, we get $\sup_{x \in (0, 1)} |f_n(x)| \rightarrow 0$. Hence $f_n \rightarrow 0$ uniformly on $(0, 1)$. $\square$

3 Lecture 2: Cauchy Criterion and Uniform Convergence Tests

3.1 Cauchy Criterion

Theorem 3.1 (Cauchy Criterion for Uniform Convergence). *A sequence f_n converges uniformly on E if and only if for every $\varepsilon > 0$ there exists N such that*

$$|f_n(x) - f_m(x)| < \varepsilon \quad \text{for all } x \in E \text{ and all } n, m \geq N.$$

Equivalently,

$$\sup_{x \in E} |f_n(x) - f_m(x)| \xrightarrow{n, m \rightarrow \infty} 0.$$

Proof. ($\Rightarrow$) If $f_n \rightarrow f$ uniformly then for given $\varepsilon > 0$ pick N so that $\sup_{x \in E} |f_k(x) - f(x)| < \varepsilon/2$ for all $k \geq N$. Then for $n, m \geq N$,

$$|f_n(x) - f_m(x)| \leq |f_n(x) - f(x)| + |f(x) - f_m(x)| < \varepsilon.$$

Taking sup over x gives the Cauchy condition.

($\Leftarrow$) Assume the Cauchy condition. Fix $x \in E$. Then $(f_n(x))$ is a Cauchy sequence in $\mathbb{R}$, so it converges to some $f(x)$. Define $f : E \rightarrow \mathbb{R}$ by this pointwise limit. Now fix $\varepsilon > 0$ and choose N such that $\sup_{x \in E} |f_n(x) - f_m(x)| < \varepsilon$ for all $n, m \geq N$. Let $m \rightarrow \infty$; since $f_m(x) \rightarrow f(x)$ for each fixed x , we get $|f_n(x) - f(x)| \leq \varepsilon$ for all $x \in E$ and $n \geq N$. Hence $\sup_{x \in E} |f_n(x) - f(x)| \leq \varepsilon$, proving uniform convergence. $\square$

3.2 Test Problems (Worked)

Example 3.2. Test convergence of $f_n(x) = \frac{x}{1 + nx^2}$ on $\mathbb{R}$.

Solution. Pointwise: for fixed x ,

$$f_n(x) = \frac{x}{1 + nx^2} \xrightarrow{n \rightarrow \infty} \begin{cases} 0, & x \neq 0, \\ 0, & x = 0, \end{cases}$$

so the pointwise limit is 0.

Uniform: compute the supremum. For $x \neq 0$, consider $g(x) = \frac{|x|}{1 + nx^2}$ for $x \geq 0$. Differentiate for $x > 0$:

$$g'(x) = \frac{(1 + nx^2) - x(2nx)}{(1 + nx^2)^2} = \frac{1 + nx^2 - 2nx^2}{(1 + nx^2)^2} = \frac{1 - nx^2}{(1 + nx^2)^2}.$$

Thus maximum occurs at $x = \frac{1}{\sqrt{n}}$ and

$$\sup_{x \in \mathbb{R}} |f_n(x)| = g\left(\frac{1}{\sqrt{n}}\right) = \frac{\frac{1}{\sqrt{n}}}{1 + n \cdot \frac{1}{n}} = \frac{1/\sqrt{n}}{2} = \frac{1}{2\sqrt{n}} \xrightarrow{n \rightarrow \infty} 0.$$

Hence $f_n \rightarrow 0$ uniformly on $\mathbb{R}$. □

Example 3.3. Test convergence of $f_n(x) = \frac{\sin(nx)}{\sqrt{n}}$ on $[0, 2\pi]$.

Solution. For each fixed x , $|\sin(nx)| \leq 1$, hence $|f_n(x)| \leq 1/\sqrt{n} \rightarrow 0$. So pointwise limit is 0.

For uniform convergence,

$$\sup_{x \in [0, 2\pi]} |f_n(x)| \leq \frac{1}{\sqrt{n}} \rightarrow 0.$$

Thus $f_n \rightarrow 0$ uniformly on $[0, 2\pi]$. □

Example 3.4. Test convergence of $f_n(x) = nxe^{-nx^2}$ on $[0, \infty)$ (and on $[0, 1]$).

Solution. Pointwise: for any fixed $x > 0$, $nx^2 \rightarrow \infty$ so $e^{-nx^2} \rightarrow 0$ faster than any power of n ; hence $nxe^{-nx^2} \rightarrow 0$. Also $f_n(0) = 0$. So pointwise limit is 0 on $[0, \infty)$.

Uniform: write $x \geq 0$ and set $t = \sqrt{n}x$. Then

$$f_n(x) = nxe^{-nx^2} = \sqrt{n}te^{-t^2}.$$

Now $\phi(t) = te^{-t^2}$ has maximum where $\phi'(t) = e^{-t^2}(1 - 2t^2) = 0$, so at $t = 1/\sqrt{2}$, and $\max_{t \geq 0} \phi(t) = \frac{1}{\sqrt{2e}}$. Thus

$$\sup_{x \geq 0} f_n(x) = \sqrt{n} \cdot \frac{1}{\sqrt{2e}} \xrightarrow{n \rightarrow \infty} \infty.$$

Therefore $f_n \not\rightarrow 0$ uniformly on $[0, \infty)$. The same supremum point occurs at $x = \frac{1}{\sqrt{2n}} \leq 1$ for large n , so also $f_n \not\rightarrow 0$ uniformly on $[0, 1]$. □

4 Lecture 3: Pointwise vs Uniform; When Pointwise $\Rightarrow$ Uniform

4.1 Classic Counterexample (Fully Worked)

Example 4.1. Define $f_n : [0, 1] \rightarrow \mathbb{R}$ by

$$f_n(x) = \begin{cases} nx, & 0 \leq x \leq \frac{1}{n}, \\ \frac{1}{n}, & \frac{1}{n} < x \leq 1. \end{cases}$$

Show $f_n \rightarrow 0$ pointwise but not uniformly on $[0, 1]$.

Solution. Fix $x \in (0, 1]$. For all large n we have $x > \frac{1}{n}$, so $f_n(x) = \frac{1}{n} \rightarrow 0$. Also $f_n(0) = 0$ for all n . Hence $f_n(x) \rightarrow 0$ for every $x \in [0, 1]$ (pointwise).

For uniform convergence, compute $\sup_{x \in [0, 1]} |f_n(x)|$. On $[0, 1/n]$, $f_n(x) = nx$ achieves maximum 1 at $x = 1/n$. Thus

$$\sup_{x \in [0, 1]} |f_n(x)| = 1 \quad \text{for all } n,$$

so $\sup_x |f_n(x) - 0| \not\rightarrow 0$. Hence the convergence is not uniform. $\square$

4.2 When does pointwise convergence become uniform?

Theorem 4.2 (Dini's Theorem). *Let K be compact, and $f_n : K \rightarrow \mathbb{R}$ continuous. Suppose $f_n \rightarrow f$ pointwise on K , f is continuous, and the convergence is monotone (either $f_n \uparrow f$ or $f_n \downarrow f$ pointwise). Then $f_n \rightarrow f$ uniformly on K .*

Proof. Assume $f_n \uparrow f$. Define $g_n = f - f_n \geq 0$. Then g_n are continuous and $g_n \downarrow 0$ pointwise. If not uniform, there exists $\varepsilon_0 > 0$ and points $x_n \in K$ with $g_n(x_n) \geq \varepsilon_0$. By compactness, pass to a subsequence $x_{n_k} \rightarrow x_* \in K$. Since g_{n_k} is decreasing in k pointwise, for any fixed m and k large enough with $n_k \geq m$,

$$g_m(x_{n_k}) \geq g_{n_k}(x_{n_k}) \geq \varepsilon_0.$$

Let $k \rightarrow \infty$; continuity of g_m gives $g_m(x_*) \geq \varepsilon_0$ for all m . But $g_m(x_*) \downarrow 0$ pointwise, contradiction. Hence uniform. $\square$

5 Lecture 4: Uniform Convergence and Differentiation

5.1 Main Theorem + Complete Proof

Theorem 5.1 (Uniform Convergence and Differentiation). *Let f_n be differentiable on $[a, b]$. Assume:*

- (1) *There exists $x_0 \in [a, b]$ such that $(f_n(x_0))$ converges in $\mathbb{R}$.*
- (2) *$f'_n \rightarrow g$ uniformly on $[a, b]$ for some function g .*

Then f_n converges uniformly on $[a, b]$ to a differentiable function f and $f' = g$ on $[a, b]$.

Proof. Fix $x \in [a, b]$. By the Fundamental Theorem of Calculus,

$$f_n(x) - f_n(x_0) = \int_{x_0}^x f'_n(t) dt.$$

Let m, n be large. Then

$$f_n(x) - f_m(x) = (f_n(x_0) - f_m(x_0)) + \int_{x_0}^x (f'_n(t) - f'_m(t)) dt.$$

Taking absolute values and using $|\int_{x_0}^x h| \leq |b - a| \sup_{[a,b]} |h|$,

$$|f_n(x) - f_m(x)| \leq |f_n(x_0) - f_m(x_0)| + (b - a) \sup_{t \in [a,b]} |f'_n(t) - f'_m(t)|.$$

Now $(f_n(x_0))$ is Cauchy, and f'_n is uniformly Cauchy (since it converges uniformly). Hence the RHS can be made $< \varepsilon$ for all $x \in [a, b]$ by choosing m, n large. Therefore f_n is uniformly Cauchy on $[a, b]$, so it converges uniformly to some f .

Next, since $f'_n \rightarrow g$ uniformly and $f_n(x) - f_n(x_0) = \int_{x_0}^x f'_n(t) dt$, take $n \rightarrow \infty$. Uniform convergence of derivatives implies

$$\int_{x_0}^x f'_n(t) dt \rightarrow \int_{x_0}^x g(t) dt$$

and uniform convergence of f_n implies $f_n(x) - f_n(x_0) \rightarrow f(x) - f(x_0)$. Thus

$$f(x) - f(x_0) = \int_{x_0}^x g(t) dt.$$

Hence f is differentiable and $f'(x) = g(x)$ for all x (by FTC). □

5.2 Application 1: $f_n(x) = \frac{\cos(nx)}{n}$

Example 5.2. Let $f_n(x) = \frac{\cos(nx)}{n}$ on $\mathbb{R}$. Show $f_n \rightarrow 0$ uniformly. Find where f'_n converges. Construct a modified example where $f_n \rightarrow 0$ uniformly but $\{f'_n\}$ is unbounded.

Solution. Uniform convergence:

$$\sup_{x \in \mathbb{R}} |f_n(x)| \leq \frac{1}{n} \rightarrow 0,$$

so $f_n \rightarrow 0$ uniformly on $\mathbb{R}$.

Derivative:

$$f'_n(x) = -\sin(nx).$$

The sequence $\sin(nx)$ converges only at points x where $\sin(nx)$ is constant in n . If $x = k\pi$ for integer k , then $\sin(nk\pi) = 0$ for all n , so $f'_n(x) \rightarrow 0$. If $x \notin \pi\mathbb{Z}$, then $\sin(nx)$ oscillates and does not converge (one can show it has at least two limit points). Thus f'_n converges exactly on $\pi\mathbb{Z}$ (to 0).

Now modify to make derivatives unbounded while $f_n \rightarrow 0$ uniformly: take

$$g_n(x) = \frac{\sin(n^2x)}{n}.$$

Then $\sup_x |g_n(x)| \leq 1/n \rightarrow 0$ so $g_n \rightarrow 0$ uniformly. But

$$g'_n(x) = n \cos(n^2 x),$$

so $\sup_x |g'_n(x)| = n \rightarrow \infty$: derivatives are unbounded. $\square$

5.3 Application 2: $f_n(x) = x - \frac{x^n}{n}$ (Correct domain discussion)

This family behaves differently depending on the domain. On all of $\mathbb{R}$ it does not even converge pointwise for $|x| > 1$, since $\frac{x^n}{n}$ diverges when $|x| > 1$.

Example 5.3. On a compact interval $[-c, c]$ with $0 < c < 1$, apply the theorem to

$$f_n(x) = x - \frac{x^n}{n}.$$

Solution. On $[-c, c]$,

$$\sup_{|x| \leq c} \left| \frac{x^n}{n} \right| \leq \frac{c^n}{n} \rightarrow 0,$$

so $f_n \rightarrow f$ uniformly where $f(x) = x$.

Differentiate:

$$f'_n(x) = 1 - x^{n-1}.$$

For $|x| \leq c < 1$,

$$\sup_{|x| \leq c} |f'_n(x) - 1| = \sup_{|x| \leq c} |x^{n-1}| \leq c^{n-1} \rightarrow 0,$$

so $f'_n \rightarrow 1$ uniformly. Also $f_n(0) = 0$ converges. Thus the differentiation theorem applies and yields $f' = 1$, consistent with $f(x) = x$. $\square$

6 Lecture 5: Uniform Convergence and Integration

6.1 Theorem + Complete Proof

Theorem 6.1 (Uniform Convergence and Integration). *Let $f_n : [a, b] \rightarrow \mathbb{R}$ be Riemann integrable for each n . If $f_n \rightarrow f$ uniformly on $[a, b]$, then f is Riemann integrable and*

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \int_a^b f_n(x) dx.$$

Proof. Uniform convergence means $\sup_{[a, b]} |f_n - f| \rightarrow 0$. Then

$$\left| \int_a^b f_n - \int_a^b f \right| \leq \int_a^b |f_n - f| \leq (b - a) \sup_{[a, b]} |f_n - f| \xrightarrow{n \rightarrow \infty} 0,$$

so $\int f_n \rightarrow \int f$. Also uniform limit of integrable functions is integrable (a standard result; can be proved using upper/lower sums). $\square$

6.2 Spikes: Why Uniform Matters (Full Computations)

Example 6.2. Define on $[0, 1]$:

$$f_n(x) = \begin{cases} n(1 - nx), & 0 < x < \frac{1}{n}, \\ 0, & \text{otherwise.} \end{cases}$$

Compute $\int_0^1 f_n$ and compare with the pointwise limit.

Solution. Pointwise, for fixed $x > 0$ and $n > \frac{1}{x}$, we have $x > \frac{1}{n}$, so $f_n(x) = 0$. Also $f_n(0) = 0$. Hence $f_n(x) \rightarrow 0$ pointwise on $[0, 1]$.

But

$$\int_0^1 f_n(x) dx = \int_0^{1/n} n(1 - nx) dx = \int_0^{1/n} (n - n^2x) dx = \left[nx - \frac{n^2x^2}{2} \right]_0^{1/n} = 1 - \frac{1}{2} = \frac{1}{2}.$$

So $\int_0^1 f_n = 1/2$ for all n , while $\int_0^1 0 = 0$. This shows *without uniform convergence* we may fail to interchange limit and integral.

Indeed, $\sup_{[0,1]} f_n = n$ (unbounded), hence no uniform convergence. $\square$

Example 6.3. Let $f_n(x) = nxe^{-nx^2}$ on $[0, 1]$. Compute $\int_0^1 f_n$ and compare with pointwise limit.

Solution. We already proved $f_n \rightarrow 0$ pointwise but not uniformly on $[0, 1]$.

Compute integral: substitute $u = nx^2$ so $du = 2nx dx$ and $nx dx = \frac{1}{2}du$:

$$\int_0^1 nxe^{-nx^2} dx = \frac{1}{2} \int_0^n e^{-u} du = \frac{1}{2}(1 - e^{-n}) \xrightarrow{n \rightarrow \infty} \frac{1}{2}.$$

So again $\lim \int f_n = 1/2 \neq 0 = \int \lim f_n$. Failure is explained by lack of uniform convergence. $\square$

7 Lecture 6: Cauchy Criterion Example $\tan^{-1}(nx)$

Example 7.1. Using the Cauchy criterion, show that $f_n(x) = \tan^{-1}(nx)$ is uniformly convergent on $[a, b]$ for $a > 0$, but not uniformly convergent on $[0, b]$.

Solution. First note pointwise behavior: for $x > 0$, $nx \rightarrow \infty$ so $\tan^{-1}(nx) \rightarrow \pi/2$. At $x = 0$, $f_n(0) = 0$. Hence the pointwise limit on $[0, b]$ is

$$f(x) = \begin{cases} 0, & x = 0, \\ \pi/2, & x \in (0, b]. \end{cases}$$

On $[a, b]$ with $a > 0$, the pointwise limit is the constant $\pi/2$.

Uniform on $[a, b]$, $a > 0$: For $x \in [a, b]$, set $y = nx \geq na$. Use the estimate (monotonicity of $\tan^{-1}$):

$$0 \leq \frac{\pi}{2} - \tan^{-1}(y) = \tan^{-1}\left(\frac{1}{y}\right) \leq \frac{1}{y}$$

(using $\tan^{-1}(u) \leq u$ for $u \geq 0$). Thus for $x \in [a, b]$,

$$0 \leq \frac{\pi}{2} - f_n(x) = \frac{\pi}{2} - \tan^{-1}(nx) \leq \frac{1}{nx} \leq \frac{1}{na}.$$

Hence

$$\sup_{x \in [a, b]} \left| f_n(x) - \frac{\pi}{2} \right| \leq \frac{1}{na} \rightarrow 0,$$

so uniform convergence holds on $[a, b]$.

Not uniform on $[0, b]$: If it were uniform on $[0, b]$, then the uniform limit would be continuous if each f_n is continuous (uniform limit of continuous functions is continuous). But the pointwise limit f has a jump discontinuity at 0. Therefore convergence cannot be uniform on $[0, b]$. $\square$

8 Lecture 7: Recursive Sequence $f_{n+1}(x) = \sqrt{x + f_n(x)}$

Example 8.1. Let $x \geq 0$ and define $f_1(x) = \sqrt{x}$, $f_{n+1}(x) = \sqrt{x + f_n(x)}$. Discuss uniform convergence on (i) $[a, b]$ with $0 < a < b$ and (ii) $[0, 1]$.

Solution. Step 1: Monotone increasing in n . Since $f_1(x) \geq 0$ for $x \geq 0$, we have $f_2(x) = \sqrt{x + f_1(x)} \geq \sqrt{x} = f_1(x)$. Assume $f_n(x) \geq f_{n-1}(x) \geq 0$. Then

$$f_{n+1}(x) = \sqrt{x + f_n(x)} \geq \sqrt{x + f_{n-1}(x)} = f_n(x),$$

so by induction $f_n(x)$ increases in n for each fixed $x \geq 0$.

Step 2: Uniform boundedness on compact intervals. Fix $x \in [0, 1]$. We claim $f_n(x) \leq 2$ for all n . Indeed, if $f_n(x) \leq 2$, then $f_{n+1}(x) = \sqrt{x + f_n(x)} \leq \sqrt{1 + 2} = \sqrt{3} < 2$. Also $f_1(x) = \sqrt{x} \leq 1 \leq 2$. Hence by induction $f_n(x) \leq 2$ on $[0, 1]$.

Similarly on $[a, b]$ one gets a uniform bound.

Step 3: Pointwise limit and formula. Since $f_n(x)$ is increasing and bounded above, it converges pointwise to some $f(x)$. Passing to the limit in $f_{n+1}(x) = \sqrt{x + f_n(x)}$ gives

$$f(x) = \sqrt{x + f(x)} \quad \Rightarrow \quad f(x)^2 - f(x) - x = 0.$$

Since $f(x) \geq 0$, we choose the positive root:

$$f(x) = \frac{1 + \sqrt{1 + 4x}}{2}.$$

This is continuous on $[0, \infty)$.

Step 4: Uniform convergence via Dini. Each f_n is continuous (composition of continuous functions). On $[a, b]$ and on $[0, 1]$ (both compact), we have:

- f_n continuous,
- $f_n \uparrow f$ pointwise,
- f continuous.

By Dini's theorem, $f_n \rightarrow f$ uniformly on $[a, b]$ (for $0 < a < b$) and also uniformly on $[0, 1]$. $\square$

9 Lecture 8: Power Series, Radius of Convergence, Abel's Theorem

9.1 Radius of Convergence

Definition 9.1 (Radius of Convergence). For a power series $\sum_{n=0}^{\infty} a_n x^n$, define

$$\alpha := \limsup_{n \rightarrow \infty} |a_n|^{1/n}.$$

Then the radius of convergence is

$$R := \begin{cases} \infty, & \alpha = 0, \\ \frac{1}{\alpha}, & 0 < \alpha < \infty, \\ 0, & \alpha = \infty. \end{cases}$$

The series converges absolutely for $|x| < R$ and diverges for $|x| > R$.

Remark 9.2. **Everywhere convergent** means $R = \infty$. **Nowhere convergent** (except possibly at 0) means $R = 0$. Examples:

$$\sum_{n=0}^{\infty} \frac{x^n}{n!} \text{ has } R = \infty, \quad \sum_{n=0}^{\infty} n! x^n \text{ has } R = 0.$$

9.2 Transforming the Radius

Theorem 9.3. If $\sum a_n x^n$ has radius of convergence R , then $\sum a_n x^{kn}$ has radius of convergence $R^{1/k}$ for each positive integer k .

Proof. Let $\alpha = \limsup |a_n|^{1/n}$, so $R = 1/\alpha$ (with conventions). Consider $\sum a_n x^{kn}$, which is $\sum b_m x^m$ where $b_{kn} = a_n$ and $b_m = 0$ otherwise. Alternatively, use root test directly:

$$\limsup_{n \rightarrow \infty} |a_n|^{1/n} |x|^k = \alpha |x|^k.$$

Convergence occurs when $\alpha |x|^k < 1$, i.e. $|x| < \alpha^{-1/k} = R^{1/k}$. $\square$

9.3 Computing Radii (Worked)

Example 9.4. Find R for $\sum_{n=1}^{\infty} \frac{(n-1)!}{n^n} x^n$.

Solution. Use $\alpha = \limsup |a_n|^{1/n}$ with $a_n = \frac{(n-1)!}{n^n}$. Using Stirling-type growth $(n!)^{1/n} \sim n/e$, we get

$$((n-1)!)^{1/n} \sim \frac{n}{e}.$$

Hence

$$|a_n|^{1/n} = \frac{((n-1)!)^{1/n}}{n} \rightarrow \frac{1}{e}.$$

So $\alpha = 1/e$ and $R = 1/\alpha = e$. □

Example 9.5. Find R for $\sum_{n=1}^{\infty} \frac{(n-1)^n}{n^n} x^n$.

Solution. Here $a_n = \left(\frac{n-1}{n}\right)^n = \left(1 - \frac{1}{n}\right)^n$. Then

$$|a_n|^{1/n} = 1 - \frac{1}{n} \rightarrow 1.$$

So $\alpha = 1$ and $R = 1$. □

9.4 Abel's Theorem (Statements) and Applications

Theorem 9.6 (Abel's Theorem: First Form (Continuity at Endpoint)). *Let $\sum_{n=0}^{\infty} a_n x^n$ have radius $R = 1$. If $\sum_{n=0}^{\infty} a_n$ converges, then*

$$\lim_{x \uparrow 1} \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} a_n.$$

Theorem 9.7 (Abel's Theorem: Second Form (Monotone Coefficients)). *If $a_n \geq 0$ and $\sum a_n$ diverges, then $\sum a_n x^n \rightarrow +\infty$ as $x \uparrow 1$. (More generally, Abel gives boundary control for power series as $x \rightarrow 1^-$.)*

Example 9.8 (Deduce $\log 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$).

Solution. For $|x| < 1$, the geometric series gives

$$\frac{1}{1+t} = \sum_{n=0}^{\infty} (-1)^n t^n.$$

Integrate from 0 to x :

$$\int_0^x \frac{1}{1+t} dt = \sum_{n=0}^{\infty} (-1)^n \int_0^x t^n dt \Rightarrow \log(1+x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1}.$$

So for $|x| < 1$,

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots.$$

Now apply Abel's theorem at $x = 1$ (the alternating harmonic series converges), giving

$$\log 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots.$$

□

Example 9.9 (Compute and prove the series for $\frac{1}{2}(\log(1+x))^2$). Show for $-1 < x \leq 1$:

$$\frac{1}{2}[\log(1+x)]^2 = \frac{x^2}{2} - \frac{x^3}{3} \left(1 + \frac{1}{2}\right) + \frac{x^4}{4} \left(1 + \frac{1}{2} + \frac{1}{3}\right) - \cdots.$$

Solution. For $|x| < 1$,

$$\log(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}.$$

Square using the Cauchy product (valid for $|x| < 1$ since the series is absolutely convergent for $|x| < 1$? Here it is not absolutely convergent at $x = 1$, but for $|x| < 1$ it is absolutely convergent indeed because $\sum |x|^n/n$ converges). Thus for $|x| < 1$,

$$(\log(1+x))^2 = \sum_{n=2}^{\infty} c_n x^n, \quad c_n = \sum_{k=1}^{n-1} (-1)^{k-1} \frac{1}{k} \cdot (-1)^{(n-k)-1} \frac{1}{n-k}.$$

Combine signs:

$$(-1)^{k-1} (-1)^{n-k-1} = (-1)^{n-2} = (-1)^n,$$

so

$$c_n = (-1)^n \sum_{k=1}^{n-1} \frac{1}{k(n-k)}.$$

Now use the identity

$$\frac{1}{k(n-k)} = \frac{1}{n} \left(\frac{1}{k} + \frac{1}{n-k} \right).$$

Therefore

$$\sum_{k=1}^{n-1} \frac{1}{k(n-k)} = \frac{1}{n} \sum_{k=1}^{n-1} \left(\frac{1}{k} + \frac{1}{n-k} \right) = \frac{1}{n} \cdot 2 \sum_{k=1}^{n-1} \frac{1}{k}.$$

Let $H_{n-1} := \sum_{k=1}^{n-1} \frac{1}{k}$ (harmonic number). Then

$$c_n = (-1)^n \cdot \frac{2}{n} H_{n-1}.$$

Hence for $|x| < 1$,

$$(\log(1+x))^2 = \sum_{n=2}^{\infty} (-1)^n \frac{2H_{n-1}}{n} x^n,$$

so

$$\frac{1}{2}(\log(1+x))^2 = \sum_{n=2}^{\infty} (-1)^n \frac{H_{n-1}}{n} x^n.$$

Writing out terms:

$$\begin{aligned} n=2: \quad (+) \frac{H_1}{2} x^2 &= \frac{1}{2} x^2, \\ n=3: \quad (-) \frac{H_2}{3} x^3 &= -\frac{1+\frac{1}{2}}{3} x^3, \\ n=4: \quad (+) \frac{H_3}{4} x^4 &= \frac{1+\frac{1}{2}+\frac{1}{3}}{4} x^4, \end{aligned}$$

and so on, which matches the required expansion.

Finally, to extend from $|x| < 1$ to $-1 < x \leq 1$, use Abel-type boundary control: the power series defines a continuous function on $(-1, 1]$ in this alternating setting, and the identity holds on $(-1, 1)$; taking $x \uparrow 1$ preserves convergence of the alternating sums, yielding the formula for $-1 < x \leq 1$. $\square$